

Arshmeen Kaur

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PROFILE

Software developer driven by results with experience in solving complex engineering problems through scalable backend systems and AI-powered solutions. Improved system performance and data accuracy by up to 30% using NLP and automation. Skilled in Python, REST APIs, and AWS, with a strong foundation in data structures and algorithms. Experienced in Agile environments, with a track record of delivering high-impact end-to-end solutions, user-focused.

EDUCATION

Lambton College

Post Graduation in Full Stack Software Development, 3.61/4

Ottawa, ON

Sept. 2024 – April 2026

Thapar Institute of Engineering and Technology

Bachelors of Engineering in Electronics and Communication, 7.33/10

India

Aug.2019 – July 2023

EXPERIENCE

Lambton College

WIL Project

January 2026 – April 2026

Ottawa, Ontario

- Built a full-stack Pet Adoption Platform to simulate real-world product delivery in an Agile environment, ensuring timely completion of weekly sprint milestones.
- Collaborated with cross-functional team members to resolve frontend-backend integration challenges and ensure seamless API communication during sprint cycles.
- Managed development tasks using Git and SDLC practices to maintain code quality, incorporate feedback from IIP sessions, and improve overall system design.

Cipher Square Technologies

Software Developer

Oct 2023 – July 2024

Delhi, India

- Developed RESTful APIs using Flask to address integration gaps between NLP modules and frontend systems, improving communication efficiency and enabling scalable deployment.
- Enhanced NLP models to address low text analysis accuracy, improving performance by 20% and streamlining user feedback collection.
- Implemented AI models to improve data extraction processes, increasing accuracy by 30% and optimizing information retrieval across applications.

Eaton Corporation

Data Science Intern

Mar. 2023 – Sep. 2023

Pune, India

- Developed a Python-based search tool using NLP techniques to address inefficient manual data retrieval from PDF and Excel files, improving search speed and usability.
- Engineered an automatic report generation tool using Python to streamline reporting workflows, integrating PyQt5 and Tkinter for improved usability.
- Automated Excel and Word workflows using openpyxl, docx, regex, and VBA macros, improving report generation efficiency by 40% and supporting audit readiness.

PROJECTS

Fur Fetch - WIL Project | *Node.js, Express, REST APIs, Full Stack*

- Built and deployed a full-stack application to digitize pet adoption processes, designing 10+ REST APIs, improving workflow efficiency and system accessibility
- Optimized backend performance by developing advanced filtering and search logic, enabling scalable and responsive data retrieval.
- Increased backend robustness by implementing input validation and error handling, ensuring 100% successful execution of CRUD operations across pets, applications, and scheduling modules.

Fringlo | *React, Typescript, Supabase, React Query, Tailwind CSS, Zod, Recharts*

- Built a group expense-sharing platform by implementing flexible bill splitting (equal/custom), automated balance calculation, and shared expense visibility, enabling real-time tracking of “who owes whom” and eliminating manual reconciliation
- Developed receipt-based expense logging with image upload (OCR-ready) by enabling shared receipt visibility across users, improving expense traceability and reducing disputes
- Designed a real-time system with history and insights using React Query and Supabase, ensuring consistent data synchronization, complete expense history, and enhanced financial transparency

APITestForge | *Python, Flask, Pytest, REST APIs, Automation Testing*

- Developed a UI-driven API testing platform to replace manual script execution, reducing test execution effort by 60% and accelerating validation cycles by 40%
- Implemented a Flask-based orchestration layer to dynamically trigger pytest suites and generate structured reports, improving test result visibility and reducing debugging time by 50%
- Deployed the application on Render, enabling live test execution and demonstrations, achieving 99% uptime and reducing setup time from hours to minutes

Spiro | *Python, Algorithm Optimization, Constraint Satisfaction, CLI Tools, Unit Testing*

- Developed a high-performance Sudoku solver using backtracking to address computational inefficiencies in traditional solving approaches.
- Optimized algorithm with smart candidate selection and preprocessing to minimize backtracking steps, reducing average solve time to under 2 seconds.
- Validated performance by testing across 1,000+ puzzles of varying difficulty levels, achieving 100% accuracy and consistent speed benchmarks.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, HTML, CSS, SQL, Bash, Shell Scripting

Technologies: REST APIs, CI/CD pipeline, Docker, DevOps

Developer Tools: GitHub, Bitbucket, Jira, Confluence, Docker, GitHub Actions, Slack, VS Code, PyCharm

Libraries: Pandas, NumPy, Matplotlib, PyQt5, Tkinter

Agile: Scrum, Sprint Planning, SDLC

Cloud Platform: AWS

Operating Systems: Linux, Ubuntu